TESTOVANIE KVALITY VNÚTORNÝCH ŠTRUKTÚR MATERIÁLOV METÓDAMI UI

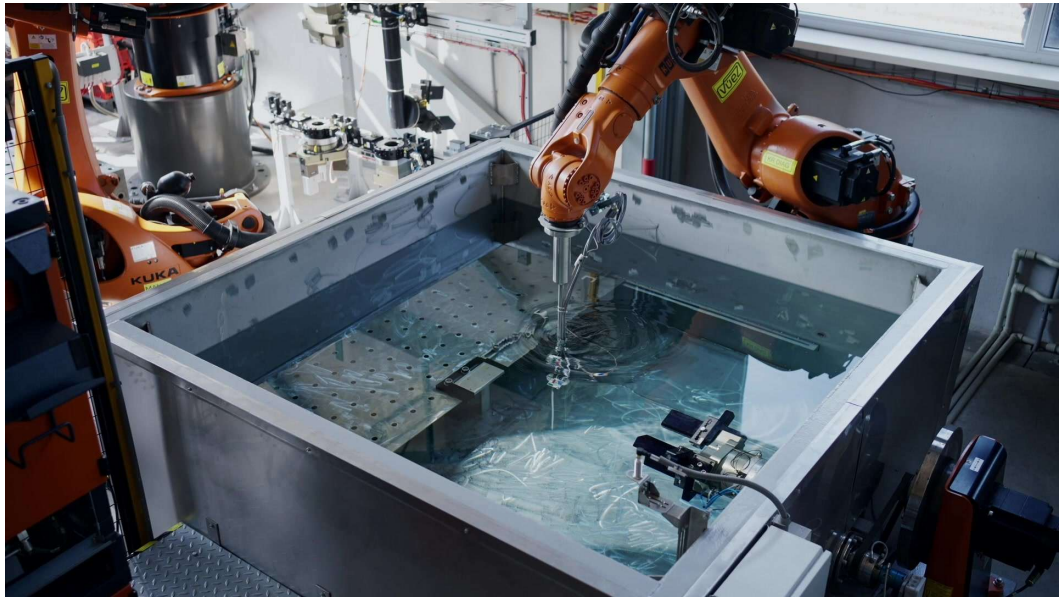
Vedúci práce : Ing. Slavomír Kajan, PhD.

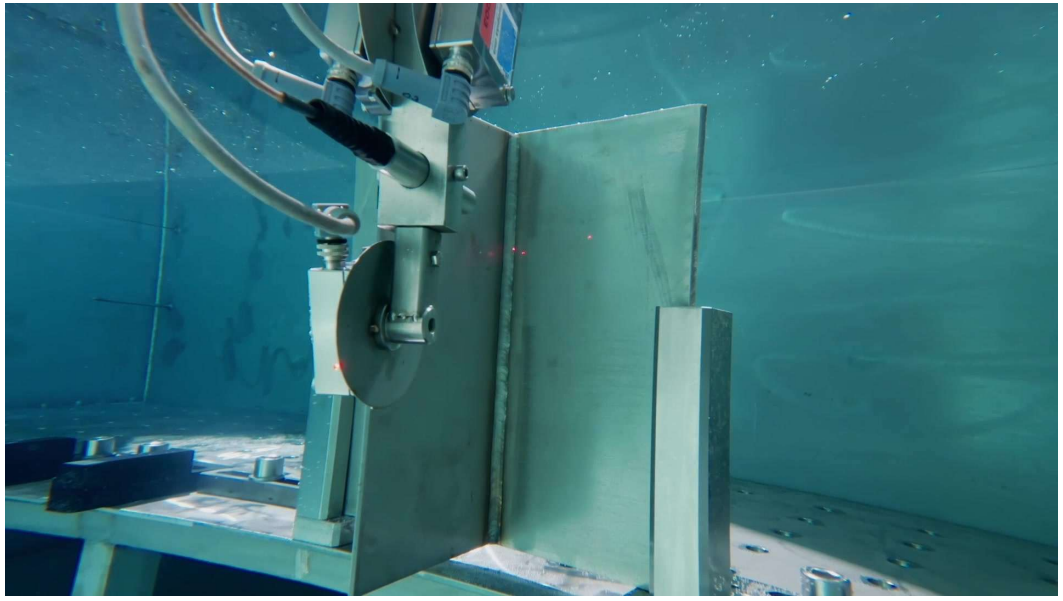
Bc. Samuel Mihálik





Robotický ultrazvuk



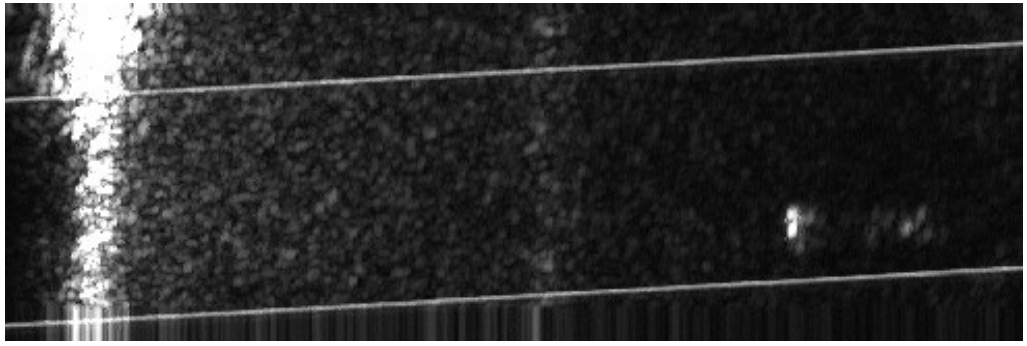


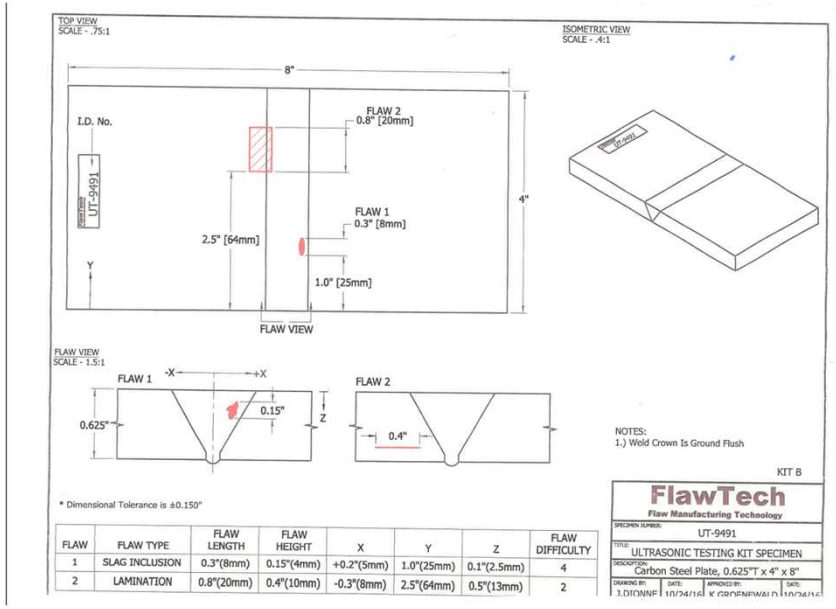
Robotický ultrazvuk

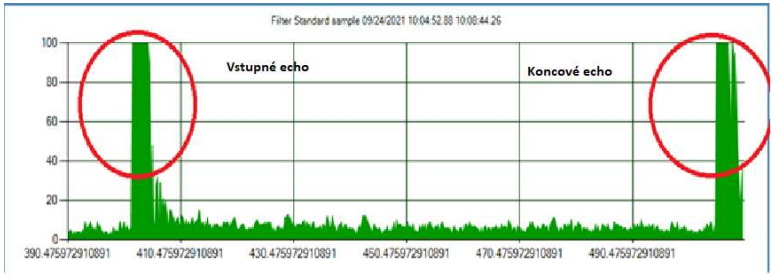
Pomocou robotického ultrazvuku získavame A-scany – sonda postupne vysiela ultrazvukové impulzy do materiálu

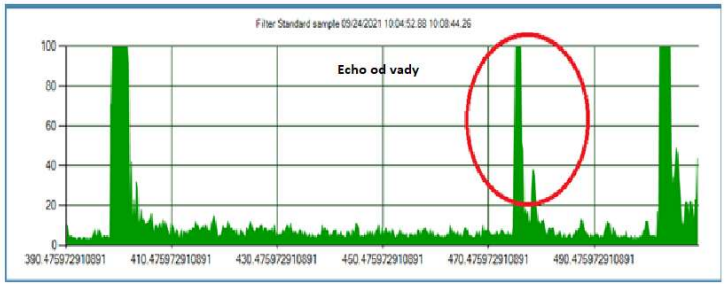
Rôzne typy defektov na oceľových platniach – 5 druhov

Viacero meraní na jednotlivé defekty – 5 meraní pre každy defekt – ručná lokalizácia defektov

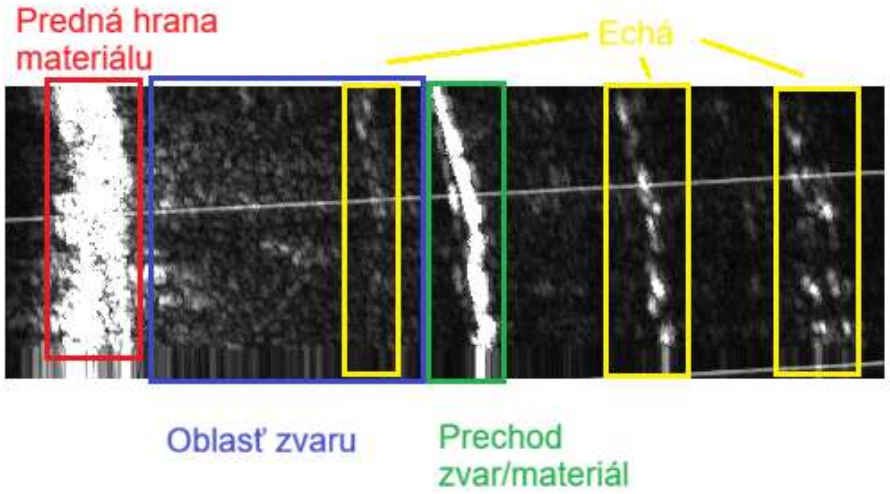


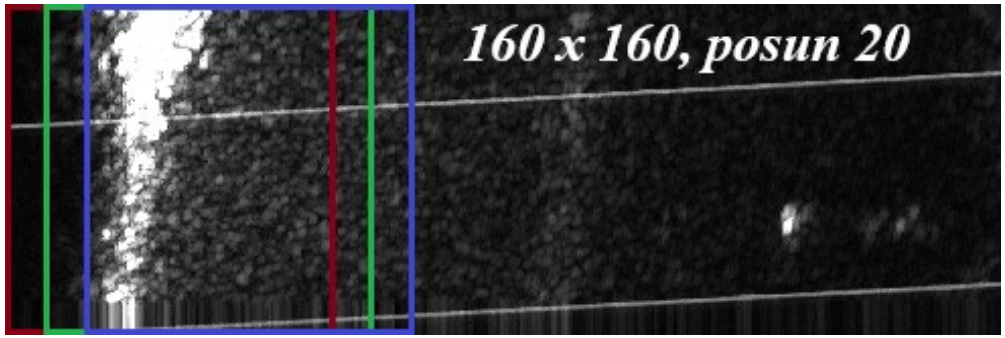


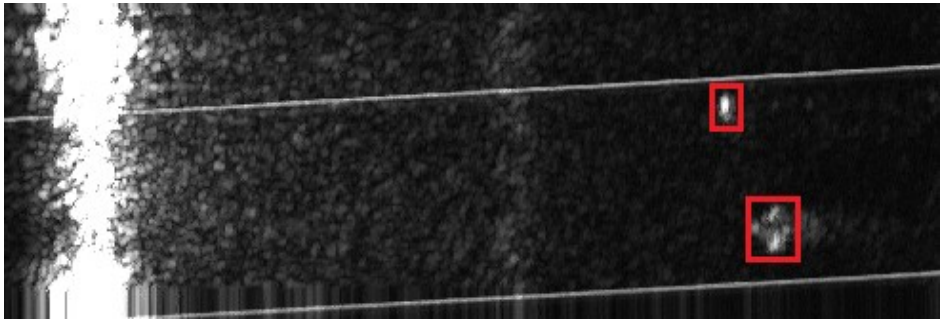


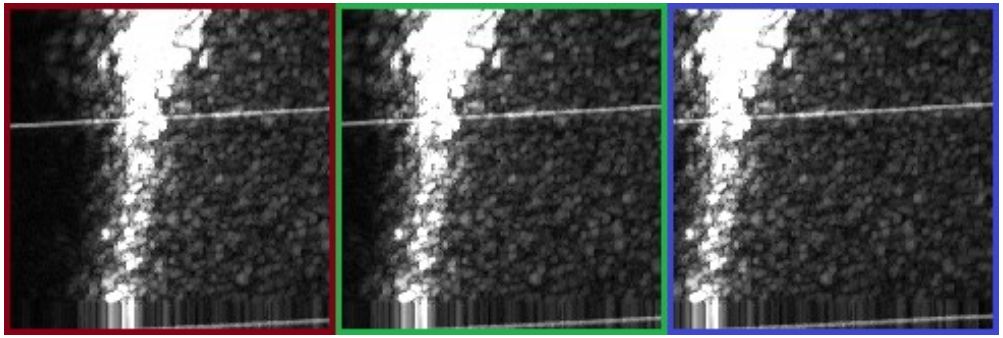


Vytvorenie Datasetu

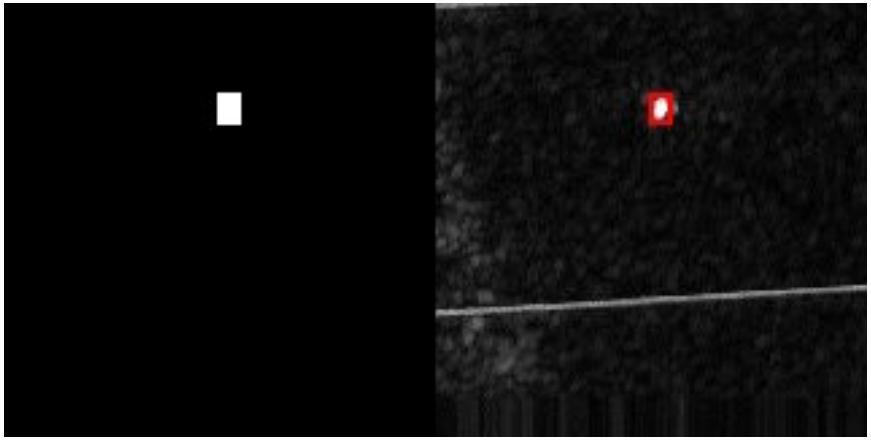








Dataset



Dataset	Počet výrezov s b1	Počet výrezov s b0
Pôvodný	3864	23 496
Balansovaný	3864	4522

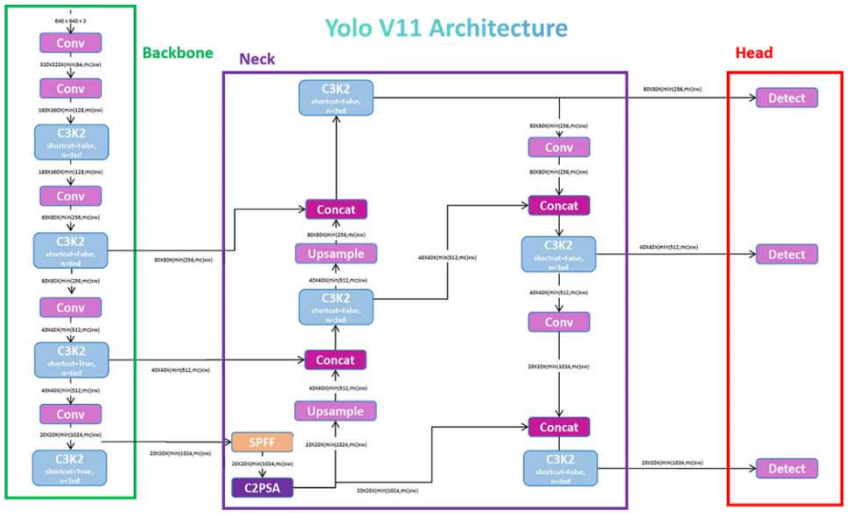
Rozdelenie datasetu:

- Trénovacia sada: UT9491–UT9500, merania 1–4 (16 meraní).
- Validačná sada: UT9491–UT9500, meranie 5 (4 merania).
- Testovacia sada: UT9498, všetky 3 merania, model ich počas trénovania nikdy nevidel.

YOLO detektor

Rozdeluje obrázok na mriežku – vrámci jedného priechodu neurónovej sieti predpovedá bounding boxy aj typy objektov. V jadre hlboká konvolučná NS.

YOLOv11 – rôzne architektúry(s,m,l,x)



Hyperparametre trénovania

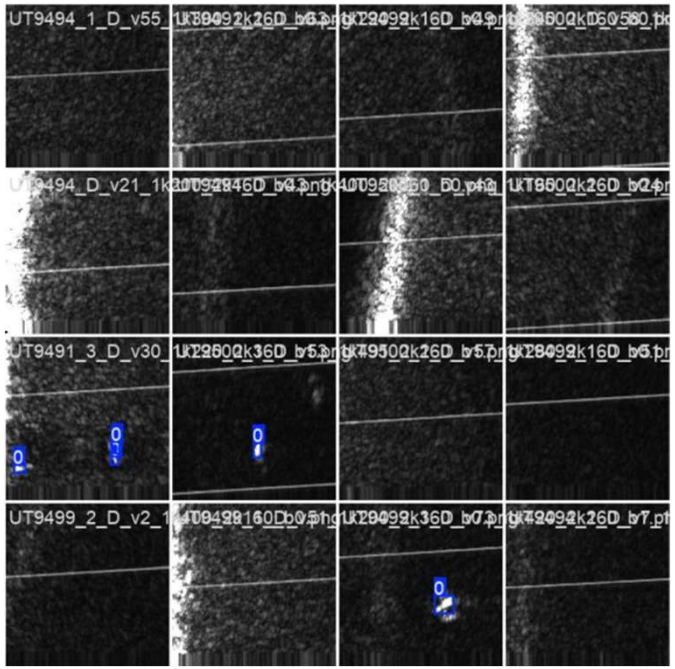
	Počet epoch	**Rozlíšenie vstupu**	**Veľkosť dávky**	**Skoré zastavenie**	**Optimalizátor (SGD)**	**Pôvodný lr**	**Momentum**
Predvolená hodnota	100	640	16	100	auto	0.001	0,9
Použitá hodnota	500	160	64	50	auto	0.001	0,9

Parameter	**Experiment 1**	**Experiment 2**	**Experiment 3**	**Experiment 4**
Model	YOLOv11n	YOLOv11n	YOLOv11s	YOLOv11s
Velkosť dávky	64	64	64	64
Optimizer	auto(SGD)	AdamW	auto(SGD)	AdamW
Počiatočný LR	auto(0.01)	0.001	auto(0.01)	0.001
Kosínusový LR	False	True	False	True
Precision	0,857	0,850	**0,873**	0,827
Recall	0,753	0,708	**0,759**	0,752
mAP50	**0,819**	0,811	0,817	0,805
val/box_loss	2,045	1,960	**1,634**	1,759

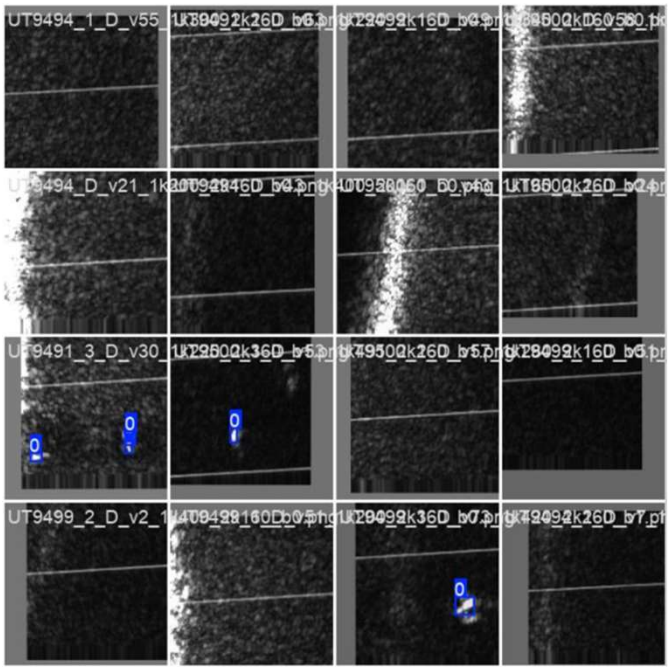
Augmentácia

Augementácia	Precision	Recall	mAP50	val/box_loss
Žiadna	0,783	0,556	0,656	1,771
Mierna	**0,873**	**0,759**	**0,817**	**1,634**
Úplná	0,831	0,712	0,788	1,643

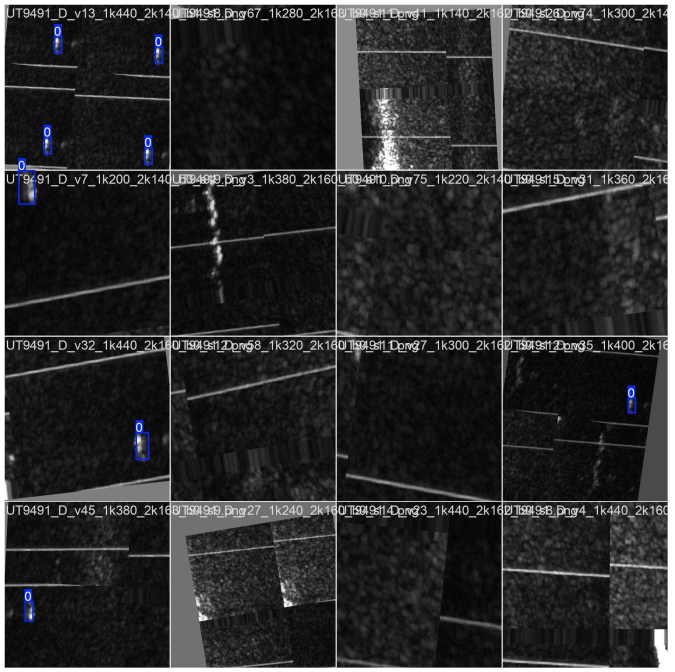
bez augmentácie



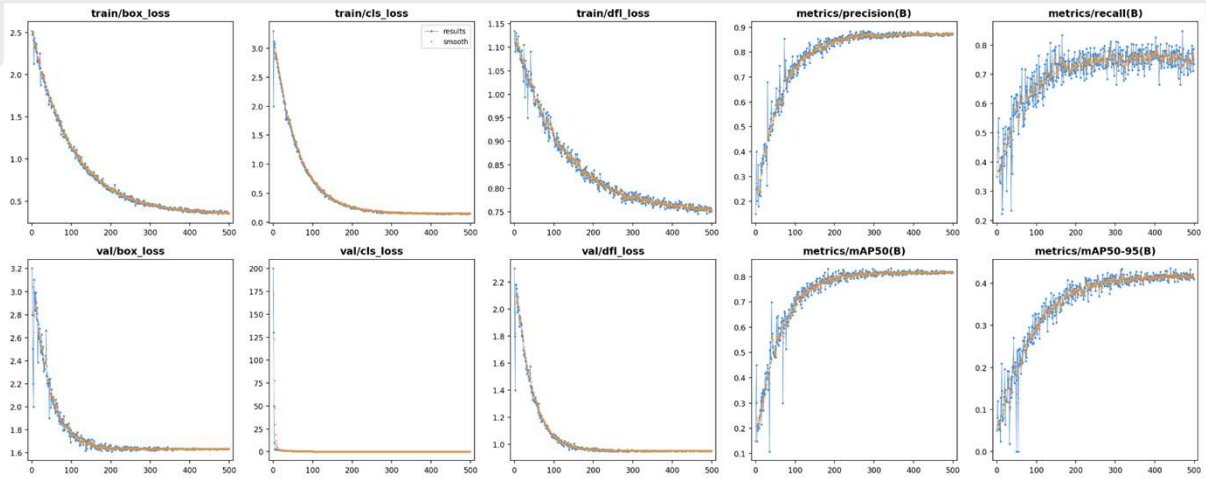
mierna

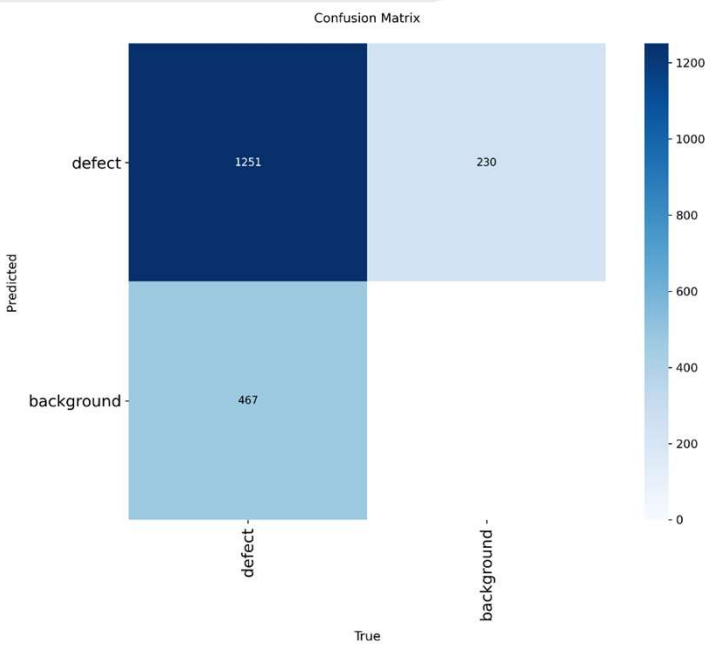


úplná



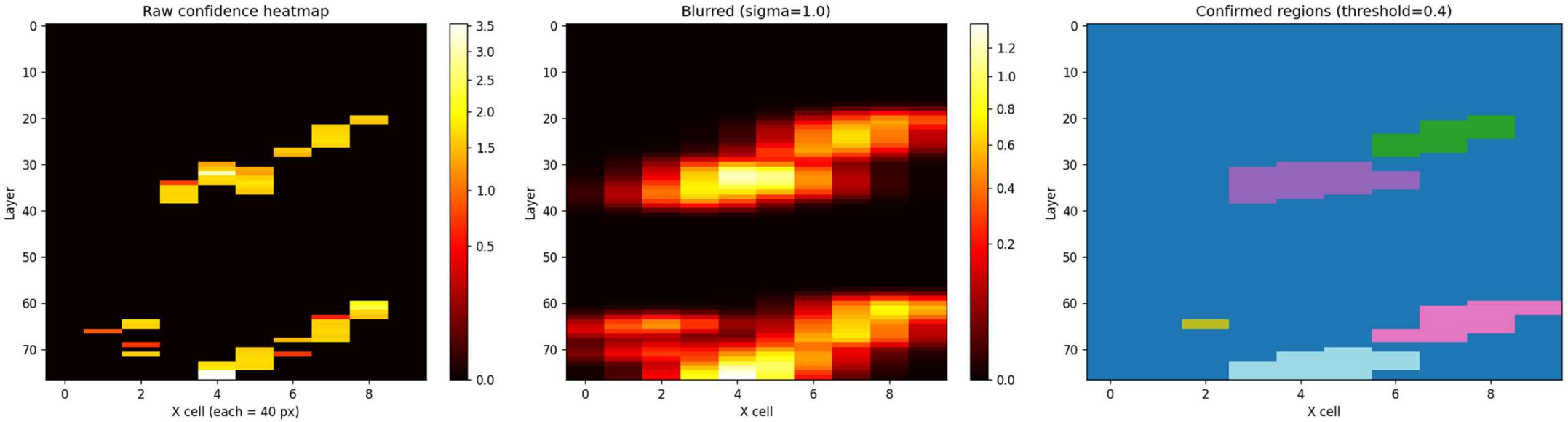
Výsledky – najlepší model





Meranie	Prah citlivosti	Detection rate [%]	Precision [%]	Accuracy [%]
UT9491_5	0,6	80,4	95,1	95,0
UT9494_5	0,6	82,6	92,2	95,4
UT9499_5	0,6	77,8	92,5	96,1
UT9500_5	0,6	90,4	96,9	97,1

Spracovanie výstupu YOLO detektora



Spracovanie výstupu YOLO detektora

```
LABEL COMPARISON REPORT (WITH HEATMAP POSTPROCESSING)
======================================================================

UT Identifier         : UT9498_3
Slice shift           : 80

Heatmap filter parameters:
  confidence_threshold : 0.5
  HEATMAP_CELL_PX      : 40 px
  HEATMAP_BLUR_SIGMA   : 1.0 cells
  HEATMAP_THRESHOLD    : 0.4

Raw detections           : 144
Confirmed defect regions : 2
Validated detections     : 83

Ground Truth:
  Defects    (b1): 45
  No-defects (b0): 335

Validated predictions (slices flagged): 44

Performance Metrics:
  True Positives  : 44 / 45
  False Negatives : 1 / 45
  False Positives : 1
  True Negatives  : 334

  Detection Rate  : 97.8%
  Precision       : 100.0%
  Overall Accuracy: 99.5%


MISSED DEFECT DIAGNOSTICS
----------------------------------------------------------------------

  Slice      : UT9498_3_D_v68_x438_y160
  Layer      : 68   X-pos: 438
  Raw hits   : 1
  Peak heat  : 0.467
  Reason     : Detection fell in a separate region or proximity check did not
cover it.
```

Meranie	**Následné spracovanie**	**Prah citlivosti**	**Falošné detekcie (FP)**	**Detection rate [%]**	**Precision [%]**	**Accuracy [%]**
UT9498_1	nie	0,6	33	**75,0**	38,9	89,5
	áno	0,6	**3**	64,3	**85,7**	**96,6**
UT9498_2	nie	0,6	40	95,6	51,8	88,9
	áno	0,6	**1**	**97,8**	**97,8**	**99,5**
UT9498_3	nie	0,6	45	**93,5**	48,9	87,4
	áno	0,6	**10**	91,3	**80,8**	**96,3**